

To suck the air out of DevLoop, Hyperloop One used a row of small pumps, housed in a metal building on one side. These can drop the pressure inside the tube to under 1/1000th of atmospheric conditions at sea level. At that point, the few air molecules left don't get in the way of a speeding vehicle. At the right hand end of the tube, one section of pipe operates as an airlock. A 3.6m (12-feet) steel disk slides across to separate that chunk from the longer tube, so that pods or other vehicles can be loaded in and out without having to pump the whole tube down to vacuum, which takes about four hours.

The company plans to run these tubes along pylons, which should be easy enough, and lets it avoid some of the engineering work that comes with laying heavy rail tracks along the ground.

Since introducing its prototype pod, Hyperloop One has completed some 200 test runs at varying speeds, collecting data on every variable it can track. In last December, it sent the pod to 386kmph (240mph) in just a few seconds—a new hyperloop record.

Robocars without steering wheels coming

General Motors (GM) has developed a car with no steering wheel. And it plans to put a fleet of these cars to work in a taxi-like service, in the US. This robo-chariot, a modified all-electric Chevrolet Bolt, doesn't have pedals either.

While Waymo, Uber and others in this space are building partnerships to deploy their autonomous technology, GM is going for vertical integration. It spent a reported \$600 million on Cruise Automation, and put the San Francisco-based startup in charge of its effort to develop fully autonomous vehicles. It bought its own lidar manufacturer—Strobe, a Pasadena-based outfit—expecting to cut sensor costs (a big problem, especially for lidar) by 99 per cent.

To lay the foundation for a business that doesn't rely on selling cars to people, it launched a car-sharing service called Maven. And it has configured its Orion assembly plant north of Detroit to build this latest generation of robocar.

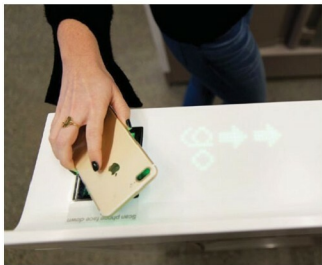


GM's latest self-driving vehicle is a modified Chevy Bolt without a steering wheel or pedals

To handle riders who demand a human touch, and to do things like call emergency services in case of a crash, GM will rely on its in-vehicle OnStar system. And, as an early test for a rideshare system, GM has built an app with which Cruise employees can call robocars for free rides around San Francisco.

Amazon opens cashierless convenience store

After more than a year of testing with an employee-only focus group, Amazon Go has finally opened to the public in Seattle, putting to the test the online retailer's technology that lets shoppers grab what they want and leave without paying a cashier. The inventory caters to health-conscious, affluent



An employee scans in to shop at the Amazon Go store (Photographer: Mike Kane/Bloomberg)

millennials rushing to their next meeting.

"We got a lot of feedback on selection," Amazon Go's technology vice president Dilip Kumar said, adding that most changes over the past year involved the store's inventory. Amazonians prefer their salad dressing on the side, for instance. And boxed meal kits to be cooked at home should be clearly labelled if they are vegan or gluten-free.

Amazon Go is the company's most ambitious effort to change the way people shop in stores. Amazon hopes the cashierless technology will help it stand out from the nation's 150,000 convenience stores where traffic jams can form at the checkout counter.

To enter the Amazon Go store, customers download a smartphone app and scan a QR code to open a glass turnstile. Those shopping in a group scan the account holder's phone once for each person entering and sensors associate them with that account. From there, machines take over, watching the items grabbed from shelves and adding them to a shopping cart. Shoppers are billed once they leave. If there are any mistakes or the customer isn't happy with an item, he just needs to push a "refund" button to have that item removed from the bill. Shoppers need not return an unwanted item to the store to get a refund!